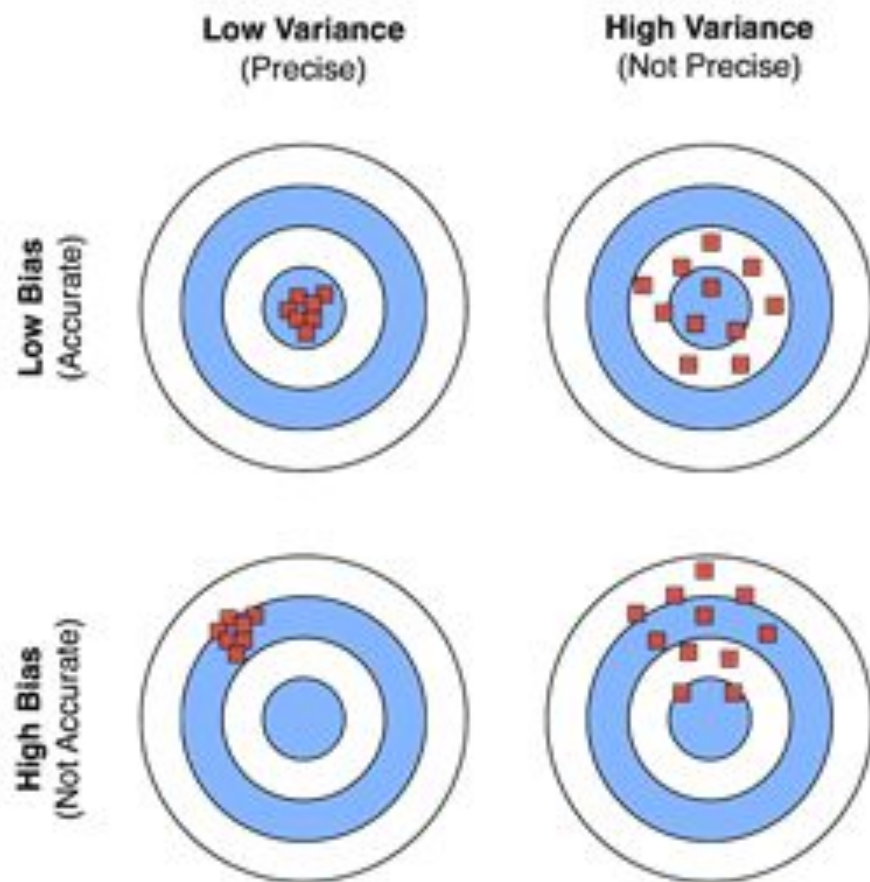
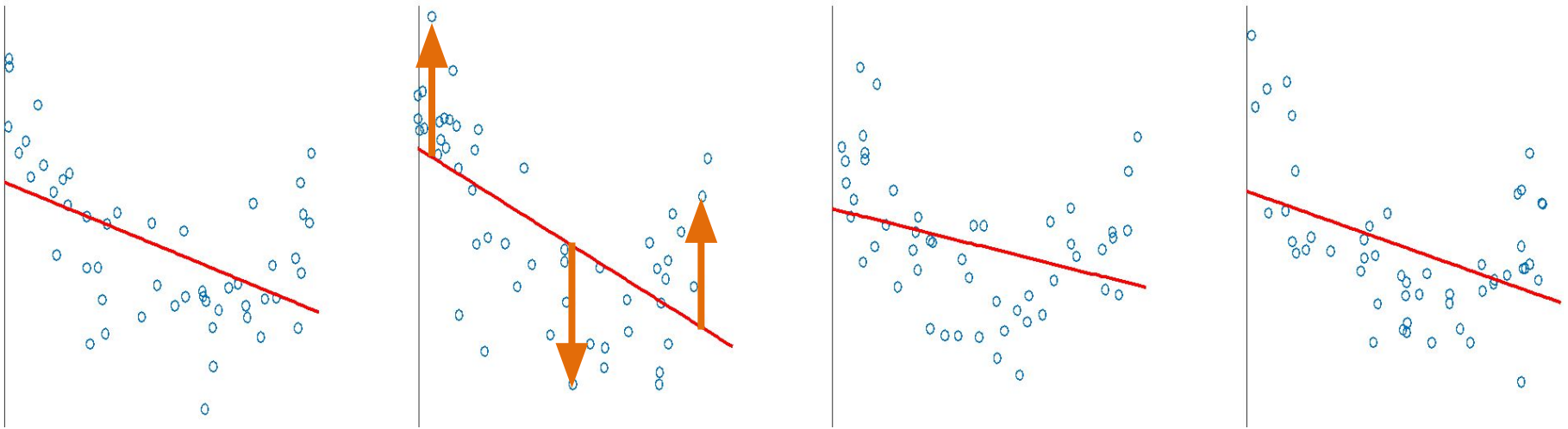


Regularization

A way to avoid overfitting



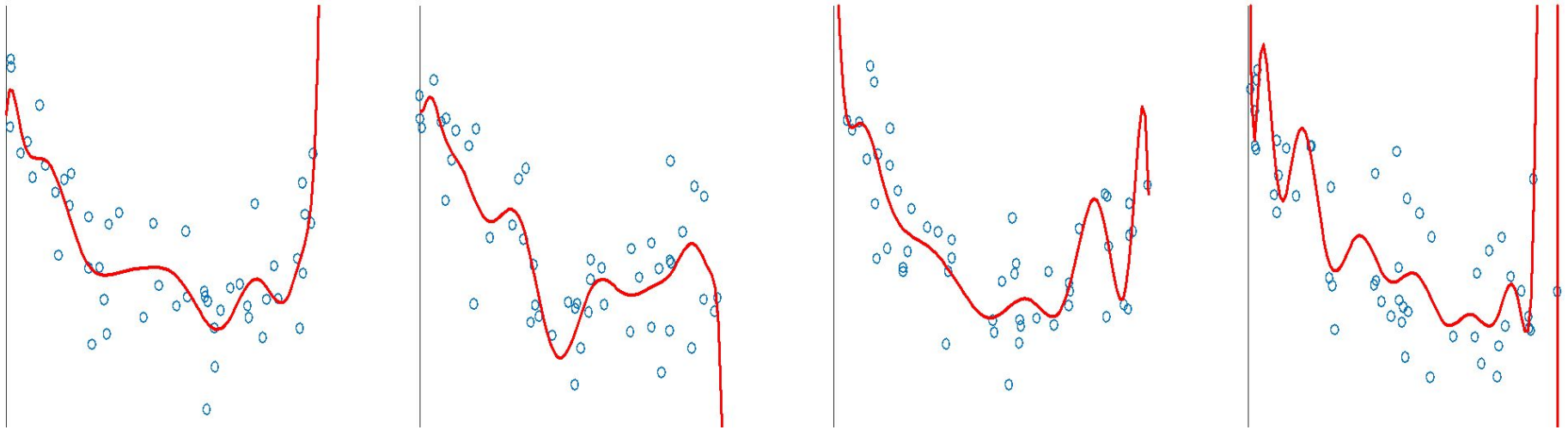
Bias



- Regardless of training sample, or size of training sample, model will produce consistent errors

Actual relationship may be quadratic

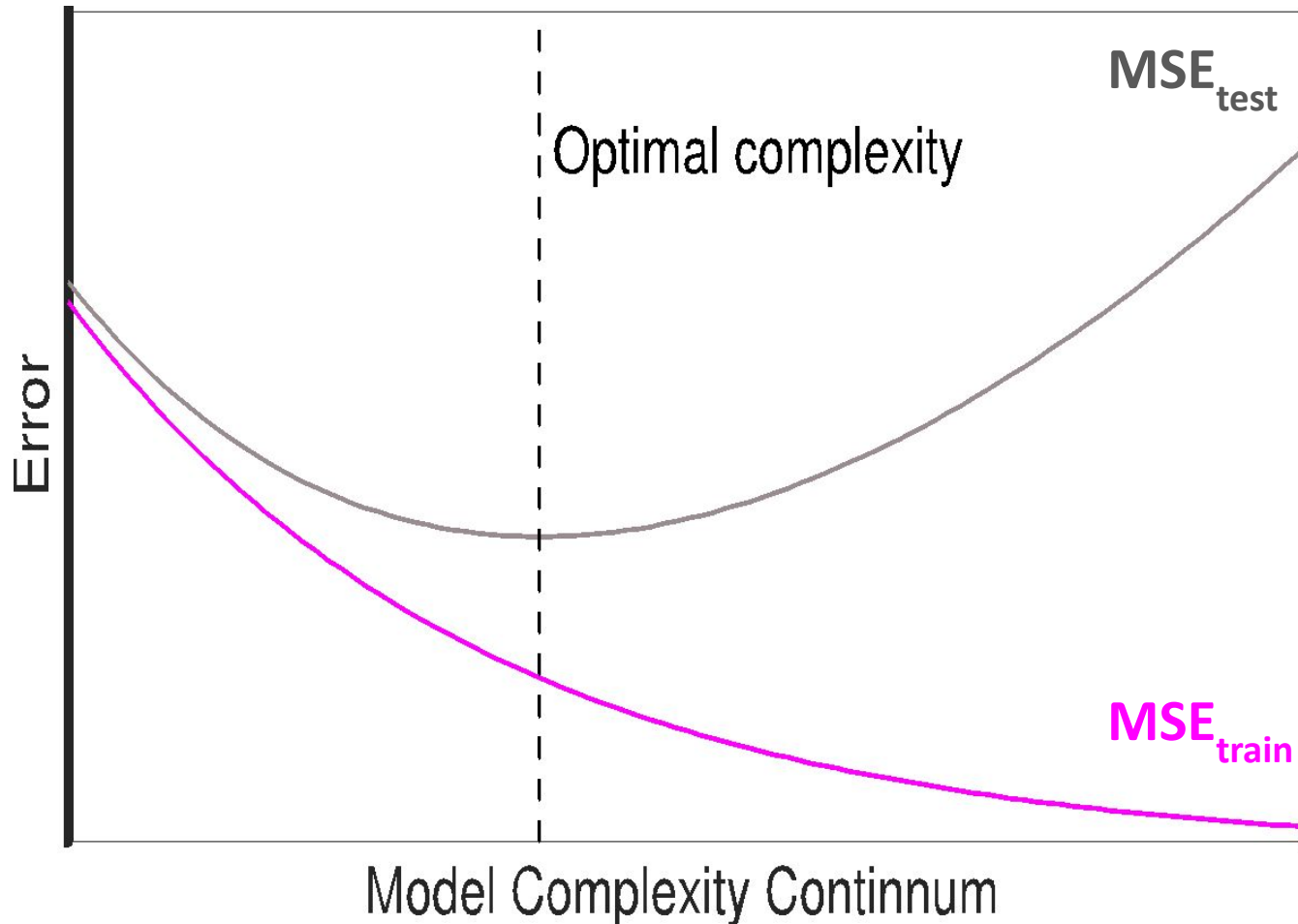
Variance



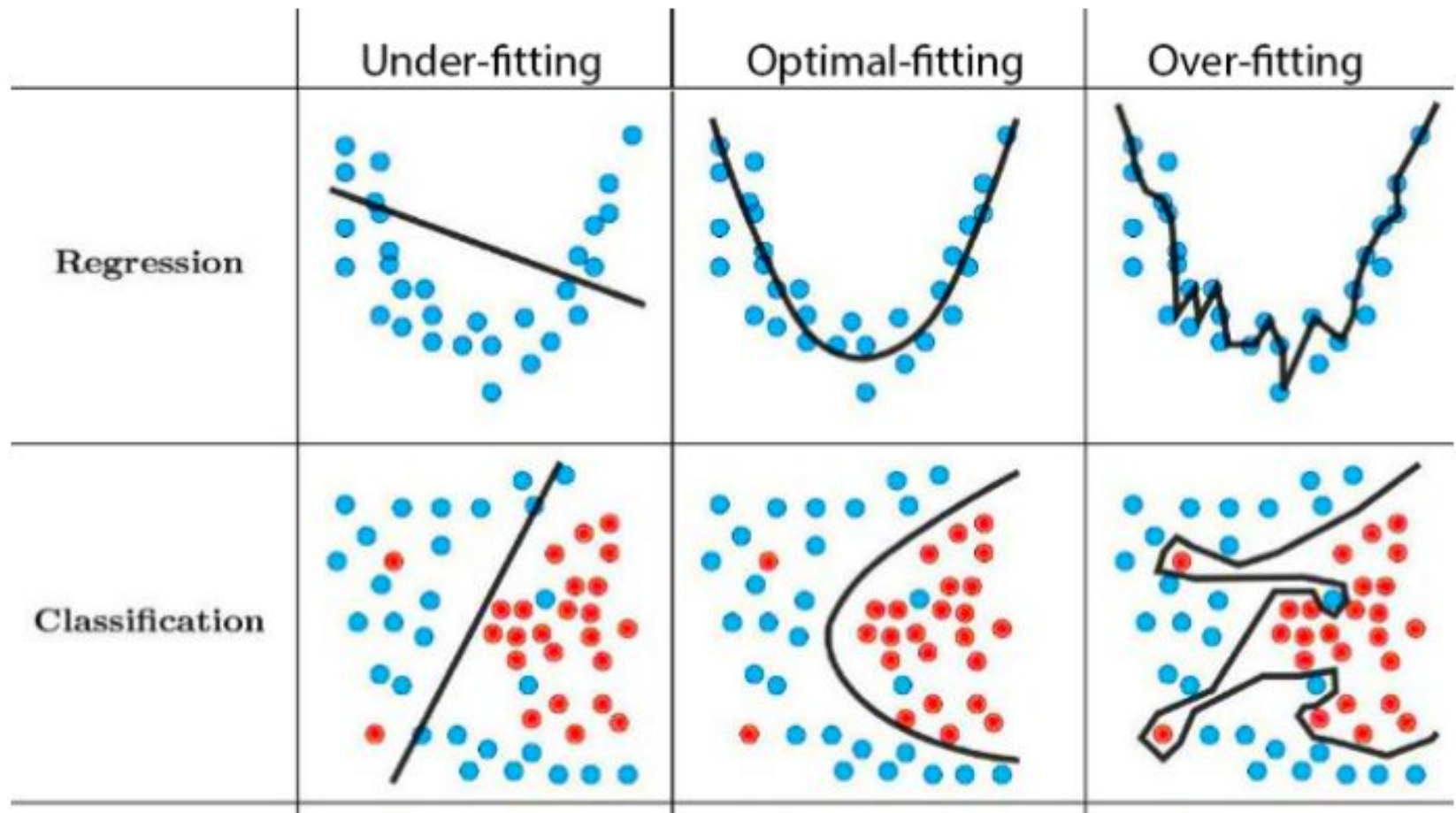
- Different samples of training data yield different model fits

We are trying to fit degree 8 polynomial

Bias-Variance Trade Off Is Revealed Via Test Set Not Training Set



Bias increases $\longleftrightarrow$ **Variance increases**

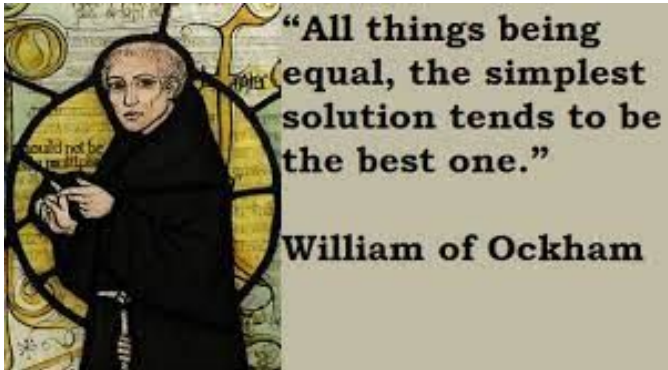


High bias ,
Low variance

Low bias ,
Low variance

Low bias ,
High variance

Regularization



- A controlled way of reducing the complexity of the fitted curve.
- We need to measure the complexity.
- We need to penalize the complex solutions.

Regularization: An Overview

- The idea of regularization revolves around modifying the criterion J ; in particular, we add a regularization term that penalizes some specified properties of the model parameters
- $J_{reg}(\Theta) = J(\Theta) + \lambda R(\Theta)$
- where λ is a scalar that gives the weight (or importance) of the regularization term.
- Fitting the model using the modified loss function J_{reg} would result in model parameters with desirable properties (specified by R).

RIDGE and LASSO Regularizations

- In Ridge regularization,
 - $J_{reg}(\Theta) = J(\Theta) + \lambda(\theta_1^2 + \dots + \theta_d^2)$
 - Note θ_0 is not used in the penalization
- In Lasso regularization,
 - $J_{reg}(\Theta) = J(\Theta) + \lambda(|\theta_1| + \dots + |\theta_d|)$
 - Here also θ_0 is not penalized.

$J(\Theta)$: SSE, Ridge regression

- $J(\Theta) =$
$$\sum_{i=1}^n \left[\left(\sum_{j=1}^d x_{ij} \theta_j + \theta_0 \right) - y_j \right]^2 \quad J_{reg}(\Theta) =$$
$$\sum_{i=1}^n \left[\left(\sum_{j=1}^d x_{ij} \theta_j + \theta_0 \right) - y_j \right]^2 +$$
$$\lambda \left(\sum_{j=1}^d \theta_j^2 \right)$$
- $\frac{\partial J_{reg}}{\partial \theta_j} =$
$$2 \left[\sum_{i=1}^n \left[\left(\sum_{j=1}^d x_{ij} \theta_j + \theta_0 \right) - y_j \right] x_{ij} + \lambda \theta_j \right]$$

This is for $j = 1$ to d . For θ_0 there is no regularization penalty (see the next slide)

Purpose of Regularization:

Regularization techniques like L1 and L2 add a penalty to the loss function, which shrinks the weights of the model. This encourages simpler models by reducing the magnitude of weights, thereby reducing the model's variance and preventing overfitting to the training data.

Role of Weights:

Weights determine how sensitive the model's output is to its inputs. Regularizing them reduces the slopes of fitted curve, which smooths the model and makes it less susceptible to noise in the data.

Role of Bias:

Bias terms, on the other hand, act as an offset or threshold, shifting the decision boundary of the model. Regularizing the bias term can result in under-fitting.

$$\bullet \frac{\partial J_{reg}}{\partial \theta_0} = 2 \left[\sum_{i=1}^n \left[\left(\sum_{j=1}^d x_{ij} \theta_j + \theta_0 \right) - y_j \right] \right]$$

$J(\Theta)$: SSE, Lasso regression

- $J(\Theta) =$
$$\sum_{i=1}^n \left[\left(\sum_{j=1}^d x_{ij} \theta_j + \theta_0 \right) - y_j \right]^2 \quad J_{reg}(\Theta) =$$
$$\sum_{i=1}^n \left[\left(\sum_{j=1}^d x_{ij} \theta_j + \theta_0 \right) - y_j \right]^2 +$$
$$\lambda \left(\sum_{j=1}^d |\theta_j| \right)$$
- This is not differentiable, hence we cannot do gradient descent,
 - Quadratic programming (using Lagrange multipliers) can be employed.

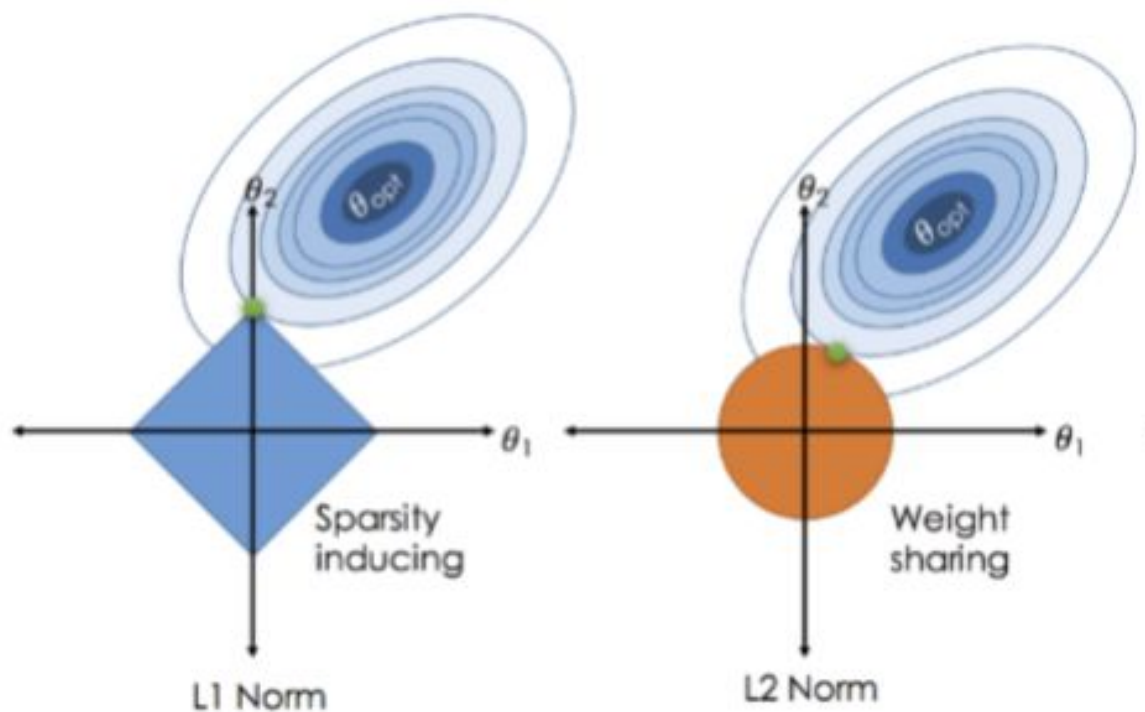
• The gradient descent update rule is

$$\theta_j(k + 1) = \theta_j(k) - \eta \frac{\partial J_{reg}}{\partial \theta_j}$$

Choosing λ

- In both ridge and LASSO regression, we see that the larger our choice of the **regularization parameter** λ , the more heavily we penalize large values in Θ ,
- If λ is close to zero, we recover the SSE, i.e. ridge and LASSO regression is just ordinary regression.
- If λ is sufficiently large, the SSE term in the regularized loss function will be insignificant and the regularization term will force $\theta_1, \dots, \theta_d$ to be close to zero. {note, θ_0 can escape this}
- To avoid ad-hoc choices, we should select λ using cross-validation.

Geometric Interpretation



Lasso regularization

Ridge regularization